MUKKA JIJNASH KUMAR

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. Portfolio: https://jijnashkumar.vercel.app

· LinkedIn: https://www.linkedin.com/in/jijnash-kumar-mukka/

· GitHub: https://github.com/Jijnash2636

Third-year B.Tech student in AI & ML at SRM Trichy (CGPA: 9.48/10), with hands-on research experience in biomedical data analysis and explainable AI. My work covers predictive models for disease detection, carbon emission control through traffic simulation, and multimodal diagnostics for neurological conditions. Recently completed an AI internship at IIT Jammu covering LLMs, Generative AI, and AI Agents. Looking for research roles where I can work on applied AI for healthcare, smart infrastructure, or sustainability.

RESEARCH INTERESTS

- Biomedical AI
- Multimodal Diagnostics
- Explainable AI (XAI)
- Clinical Decision Systems
- Predictive Modelling

EDUCATION

B.Tech, Artificial intelligence & machine learning (2023 – Present)
SRM Institute of Science and Technology, Tiruchirappalli, CGPA – 9.48/10
Graduation year (2027)

12th (Senior Secondary Examination) 2022
STATE Board, 76.8%

10th (Secondary Examination) 2020
STATE Board, 99%

Research Experience

IIIT Nagpur – Research Intern (CSE)
Dec 2025 – Feb 2026
• Developed multimodal ML models for Parkinson's disease detection using voice, facial, and hand biomarkers
• Designed pipelines for non-imaging biomedical data
• Explored feature fusion techniques for improved prediction
• Evaluated models using standard classification metrics

Research & Projects

EcoSignal-CP: Predictive Carbon Emission Control
- Built CO_2 prediction model using SUMO; reduced peak emissions by ~11.8%
- Applied simulation-based optimization

AI-Based Clinical Blood Analysis (CBC)
- Developed ML model for disease prediction using hematological data
- Performed feature engineering and evaluation (ROC-AUC)

Multimodal Parkinson's Detection System
- Developed ML model using voice, facial, and motor biomarkers
- Focused on early, non-invasive detection

Emergency Brain 360 – AI Clinical Triage (Top 6 National)

- Built AI-assisted triage system with real-time dashboard

Hospital Analytics Platform (MedAIthon – 5th Place)

- Designed system for patient flow optimization

Technical Skills

Python, C++

Machine Learning, Feature Engineering, Explainable AI

Scikit-learn, Pandas, NumPy

SUMO, FastAPI, Google Cloud (Vertex AI)

Research & Publications

Research Paper (In Progress): *EcoSignal-CP: Carbon Pressure Control for Urban CO_2 Emission Stabilization*

Book Chapter Publication – The Dawn Publishers (2024)

Internships & Training

AI Internship: LLMs, GenAI & AI Agents - IIT Jammu (Winter School) | Dec 2025 – Mar 2026 |
Cert ID: TEC/IIT/WS25/C/M2/0006
Generative AI Internship - Google Cloud (AICTE & EduSkills)
AI & Machine Learning Internship - AWS Academy
ServiceNow AI Automation Internship - AICTE & SmartBridge
AI Agent for Healthcare Diagnostics - Infosys Springboard

Achievements

- Top 6 National Rank - Smartathon 2025
- 5th Place & Special Prize – MedAIthon 2025
- CGPA 9.48/10 – Consistent Academic Excellence

CERTIFICATIONS

- **RHCSA** (Red Hat Certified System Administrator)
- Google Cloud GenAI | AWS ML | ServiceNow AI | Infosys AI | IIT Jammu AI (LLMs, GenAI, AI Agents)

Professional Objective

To pursue advanced research in AI-driven interdisciplinary systems, contributing to scalable solutions in healthcare diagnostics, smart urban infrastructure, and carbon emission control, while preparing for graduate-level research and innovation.

REFERENCES

Dr. Jitendra V. Tembhurne
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Department of Computer Science & Engineering
Indian Institute of Information Technology, Nagpur
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Dr. Balaji Ganesh Rajagopal
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